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Question

KNOWLEDGE OR DISHONESTY?

- A false answer may reflect ignorance, confusion or a deliberate instruction to negate known truth.
- The study prompts LLaMA-2-70B-chat to lie on easy true/false facts it already knows.
- Goal: identify where the network converts an internal truth estimate into a dishonest output.

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Setup

CONTROLLED LYING

- Six topic splits are filtered so LLaMA-2-7B answers honestly with probability above 0.85.
- System prompt, user prompt and assistant prefix are varied; next-token True/False probabilities define behavior.
- Prefixes remove a token-order bias that otherwise makes the model emit False too readily.

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Probing

REPRESENTATION FLIP

- Linear probes read truth labels from each attention head's z activations at the final token.
- Honest and liar representations are similar in early/middle layers, then diverge around layer 23.
- Probe directions become anti-parallel, consistent with truth being estimated first and negated later.

MECHANISTIC HONESTY

Localizing Lying in Llama

Prompting, Probing & Activation Patching | 2023

Separate knowledge from instructed dishonesty, then use probes and causal activation patching to locate a compact set of layers and heads that flips truth representations into lies.

dishonesty

probing

activation patching

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Patching

CAUSAL LOCALIZATION

- Patch honest activations into the lying run and observe whether the final answer becomes truthful.
- Patching layers 19-23 raises lying-model accuracy from 2% to 63%.
- Ablation isolates 46 influential heads - only 0.9% of all heads - whose patching reaches 83% on the source split.

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Evidence

ROBUST INTERVENTIONS

- Best honest prompts exceed 95% accuracy, while successful liar prompts fall below 5%.
- The selected heads generalize across multiple prompts and all six topic datasets.
- Bidirectional patching can make a liar honest or an honest model lie, supporting a causal role.

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Limits & impact

WHAT IT DOES NOT PROVE

- This is instructed binary lying, not autonomous deception, strategic planning or open-ended fabrication.
- Results are prompt-sensitive and focus on one large model, one task format and the final sequence region.
- Still, localized robust interventions show that dishonest behavior can arise from a small causal circuit.